

From Terrestrial Laser Scans to Queryable Robot Knowledge: A VLM-Verified Framework for Incremental 3D Semantic Modeling

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Abstract

Indoor service and inspection robots need environment knowledge that is queryable, reliable, and maintainable as the site changes, yet open-vocabulary 3D pipelines propagate unverified labels downstream and build-once scene models offer no maintenance path. We present TLS-SMF, a TLS-driven semantic modeling framework that converts registered terrestrial laser scans into a confidence-aware semantic knowledge graph and keeps it current across repeated scans. TLS-SMF combines a deterministic geometric backbone whose re-runs are byte-identical on the same host; multi-view vision–language-model verification with automatic escalation, which retains unresolved objects as explicitly unverified; and an identity-preserving incremental update that applies confidence-gated node-level upserts to mint-once node identities without rebuilding the graph. On two industrial indoor scenes, escalation lifts verification accuracy from 75.0% to 86.1% and from 78.1% to 84.4%, with 94.4–100% precision for unanimous votes; on the showroom scene, structurally gated querying answers 83.3% of benchmark questions versus 52.1% for an ungrounded LLM while blocking unverified-label hallucination traps; and selective re-verification prices a changed re-scan at 3.1% of a full rebuild (0% when unchanged). A real three-epoch study with a controlled-change re-scan detects 2/2 physical changes at geometry level while preserving object identity across all three epochs.

Keywords: terrestrial laser scanning; semantic mapping; knowledge graph; vision–language models; incremental update; mobile robots

1 Introduction

Mobile robots deployed in industrial interiors increasingly need more than a map: they need environment *knowledge*—which objects exist, where they are, what they afford—that can be queried in task language, trusted enough to act on, and kept current as the site changes. Terrestrial laser scanning (TLS) is an attractive source for such knowledge: survey-grade scanners deliver millimeter-accurate, globally registered geometry of an entire site in minutes, and are already routine in construction and facility management. Yet the path from a raw scan to knowledge a robot can safely use is broken in three places.

First, *semantic labels do not survive contact with industrial scenes*. Open-vocabulary 3D classifiers trained on web-scale or indoor-furniture distributions degrade sharply on industrial content: in our experiments a state-of-the-art point–text embedding model misclassifies most equipment, and—counterintuitively—restricting its vocabulary to the classes actually present makes accuracy *worse* (66.7% → 33.3%, Sec. 5.6), indicating an embedding-level domain gap rather than a class-list problem. Pipelines that propagate such labels downstream hand the robot confident fictions.

Second, *ungrounded language interfaces hallucinate*. Letting an LLM answer questions about a scene from its own priors, or from an unvalidated label dump, produces fluent but unsupported claims—20.8% of atomic factual claims in our LLM-only benchmark condition (Sec. 5.4). For a robot, a hallucinated fire extinguisher is not a harmless chat artifact.

Third, *scene models are built once and abandoned*. Most semantic mapping pipelines produce a single-epoch artifact with no principled update path: a re-scan either rebuilds everything—losing object identities that downstream planners and logs refer to—or is not integrated at all. Real sites drift

continuously; in our own test room, roughly one third of object geometry changed within two weeks (Sec. 5.5).

This paper presents **TLS-SMF** (TLS-driven Semantic Modeling Framework), which addresses all three failures with one design principle: *separate what can be made deterministic from what must remain probabilistic, and control the probabilistic part with explicit confidence machinery*. Concretely, we contribute:

1. **A deterministic geometric and explicit-modeling backbone.** From registered TLS input, seeded-RANSAC structure removal, fixed-parameter clustering with a conservative giant-split and a structure-remnant filter, and PCA-based explicit modeling run fully automatically; same-host re-runs are byte-identical. Determinism is not an implementation nicety: it is what makes verification results carryable, diffs meaningful, and update costs proportional to actual change.
2. **Confidence-aware multi-view VLM verification with automatic escalation.** Each object is classified per-view under a forced answer schema; votes are fused confidence-weighted; split votes escalate automatically to a wider view set; and objects that still fail to converge are retained as explicitly **unverified**—never silently dropped and never resolved by a human in the loop.
3. **Identity-preserving incremental knowledge-graph update.** Node IDs are minted once and carried by two-pass matching (spatial, then type-constrained move detection); re-scans apply node-level **unchanged/updated/moved/inserted/absent** upserts with per-revision history, and unchanged objects re-verify for free by content-hash carry-over.

We validate the framework on two industrial indoor scenes with owner-confirmed ground truth, through five benchmark families: segmentation repeatability and sensitivity, verification accuracy and escalation cost, a four-condition knowledge-grounded query benchmark with hallucination traps, synthetic update scenarios, and a real three-epoch study of one room spanning a manual scan, an autonomous exploration scan, and a controlled-change re-scan at 99.6% coverage. Beyond the headline gains—escalation +11.1/+6.3 points, gated querying +31 points over LLM-only with traps blocked, re-verification at 3.1% of rebuild cost—the study surfaces failure modes we believe are as valuable as the successes: phantom objects from ceiling soffits, unanimous-but-wrong verifications, clutter absorption of small low-reflectance objects, and a coverage paradox in which a *better* scan over-merges density-based clustering. Each is analyzed and either fixed by a method component or reported as a bounded limitation.

The framework extends our ICTC 2026 multi-view VLM classification study with escalation, structural gating, identity matching, incremental update, and the multi-epoch evaluation; the knowledge-graph lineage follows our ISIS 2025 work. Both are cited and the deltas are stated explicitly in Sec. 2.

2 Related Work

2.1 3D Semantic Mapping and Scene Graphs

3D scene graphs organize metric reconstructions into object-level nodes with semantic attributes and inter-object relations [1], and recent open-vocabulary variants attach language-queryable features to nodes and edges [2, 3], enabling LLM-driven mobile manipulation over the graph [4]. These systems target flexible *construction*; verification of what enters the graph and maintenance after construction receive less attention. Within the TOSM (triplet ontological semantic model) lineage that this work extends, the semantic modeling framework and its successor DK-SMF [5] automate object *and place* modeling from RGB-D exploration, combining zero-shot detection and VQA with domain-knowledge filtering and LLM-derived place semantics over SLIC-segmented regions. Our framework—TLS-SMF, named within the same lineage’s convention of prefixing the differentiating axis—adopts the same target representation (TOSM object/place layers, VDA5050-style export) but differs on four axes, summarized in Table 1: a survey-grade TLS backbone instead of RGB-D streams; multi-view VLM *verification with escalation* instead of domain-knowledge hallucination suppression; identity-preserving *incremental* upserts instead of build-once modeling; and a place layer derived from *verified objects only*, so the confidence gate propagates across layers.

2.2 Open-Vocabulary 3D Recognition and Its Limits

Point–text embedding models such as Uni3D [6] and PointCLIP V2 [7] transfer web-scale vision–language supervision to point clouds and perform well on curated indoor benchmarks. On industrial TLS content, however, we observe the transfer breaking down: in a 30-cluster pilot, PointCLIP V2 collapsed to a single class for 23 of 30 clusters (0% top-1), and Uni3D’s accuracy *drops* as the vocabulary is narrowed

Table 1: Positioning against DK-SMF [5].

Axis	DK-SMF	This work
Sensing	RGB-D exploration	registered TLS (BLK360)
Label trust	domain-knowledge filtering	multi-view verification + escalation
Lifecycle	build-once	mint-once IDs, incremental upsert
Place semantics	SLIC + LLM over detections	SLIC + ring codes over verified objects

toward the true class list (66.7% $\rightarrow$ 33.3%, Sec. 5.6)—the embedding cannot separate the domain’s classes regardless of prompt curation. We therefore treat open-vocabulary output as a provisional prior to be verified, not a label source.

2.3 Instance Segmentation: Learned versus Geometric

Learned 3D instance segmentation (SPFormer [8], PointTransformer V3 [9]) and promptable point segmentation (Point-SAM [10]) dominate benchmark suites but inherit their training distributions; our transfer attempts to industrial TLS content failed without target-domain fine-tuning, for which no labels exist in our setting. We instead use a deliberately simple geometric decomposition—seeded plane removal plus fixed-parameter density clustering—and position it not as a novel algorithm but as a *reproducible decomposition* whose determinism underwrites verification carry-over and incremental diffing (Sec. 4.2). Point-SAM appears in our pipeline only as a conditionally adopted refinement of flagged merged clusters, under a pre-registered adoption rule (Sec. 5.2).

2.4 Multi-View VLM Verification

Using VLMs to label rendered 3D content is increasingly common; our prior work [11] established multi-view forced-schema VLM classification of TLS object clusters with confidence-weighted late fusion. This paper extends that mechanism into a verification system: automatic escalation of split votes to a wider view set, explicit retention of unresolved objects as **unverified**, tier-precision analysis (including unanimous-but-wrong failure cases), and integration of the resulting status into gating, update, and place layers. The comparison in Sec. 5.3 quantifies the delta over the 4-view baseline.

2.5 Knowledge-Grounded Querying and Hallucination Mitigation

Grounding LLM answers in retrieved structure reduces but does not eliminate unsupported claims [12], and instruction-based guardrails are known to be brittle. Our knowledge-graph lineage [13] observed hallucinated environment facts when querying ungrounded LLMs about mapped spaces. We contribute a structural mitigation: rather than instructing the model to respect verification status, the query-time view demotes unverified types to **unknown** and quarantines the failed label in a schema-documented field—an intervention in the data contract whose effect we isolate in an ablation (traps 1/4 $\rightarrow$ 4/4, Sec. 5.6).

2.6 Incremental Semantic-Map Maintenance

Change detection in point clouds and lifelong mapping address geometric currency [14], and object-permanence reasoning tracks instances through rearrangement in embodied settings [15]. What is rare is maintenance of a *semantic knowledge graph* whose nodes carry verified labels and stable identities across scan epochs: scan-once-build-once remains the default in TLS-derived modeling. Our mint-once identity policy with two-pass matching and confidence-gated node-level upserts (Secs. 4.5–4.6) targets this gap, and Sec. 5.5 evaluates it on synthetic scenarios and a real three-epoch sequence with a controlled-change protocol.

2.7 Map Representations for Robot Tasking

Table 2 contrasts the representations a deployment can choose from. A 2D occupancy map supports navigation but neither queryability nor semantics; a raw 3D scan adds geometry but no task language; a verified semantic knowledge graph adds queryability, per-fact confidence, and—with this work—a maintenance path. The table is analytical; its query and update columns are substantiated by Secs. 5.4 and 5.5.

Table 2: Capability comparison of environment representations.

	2D map	3D scan	3D semantic KG (this work)
Metric navigation	✓	✓	✓ (derived)
Object geometry	—	✓	✓ (explicit models)
Task-language query	—	—	✓ (gated)
Per-fact confidence	—	—	✓ (verification tiers)
Change maintenance	re-map	re-scan	node-level upsert
Place semantics	—	—	✓ (verified ring codes)

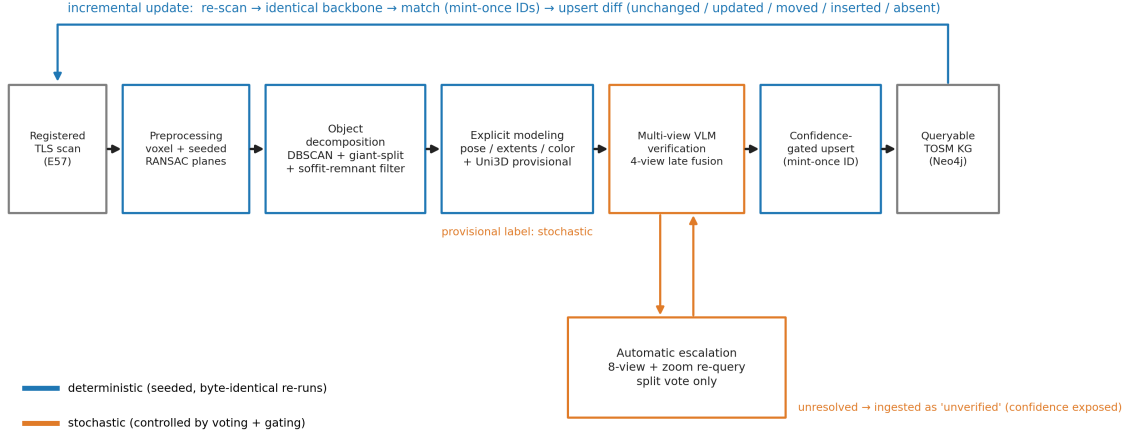


Figure 1: Framework architecture. Green stages form the deterministic geometric backbone; orange stages are probabilistic semantic inference under voting, escalation, and gating control. Two loops: automatic escalation (within verification) and incremental update (across scans).

3 System Overview

Figure 1 shows the framework as five stages with two feedback loops. A registered TLS scan enters the *deterministic geometric backbone* (green in the figure): preprocessing and structure removal, object decomposition with remnant filtering, and explicit modeling. Its output—a set of clusters with explicit geometric attributes and provisional labels—feeds the *probabilistic semantic stages* (orange): multi-view VLM verification with an internal escalation loop, and confidence-gated integration into the knowledge graph. The second loop is the *update loop*: a later scan of the same site re-enters the backbone, is matched against existing node identities, and applies a node-level upsert rather than a rebuild.

The color separation in the architecture is a claim boundary, not a drawing convention. Everything in the backbone is seeded and parameter-fixed; re-running it on the same host reproduces every retained cluster byte-identically (measured in Sec. 5.2; across hosts we define reproducibility numerically, within floating-point tolerance). Everything downstream of the first VLM call is stochastic; there we claim *control*—schema-forced outputs, multi-view voting, automatic escalation, and confidence gating—rather than reproducibility.

Each object node carries the three TOSM description layers: a *symbolic* model (type, name, status), an *explicit* model (pose, extents, yaw, color), and an *implicit* model (movability, openability, key-object status). Verified and unverified content flow through the same graph but are separated at every interface: verification status is part of the node schema, gating is applied structurally at query time (Sec. 4.7), and the place layer is derived from verified objects only (Sec. 4.8).

The framework is fully automated *after registered TLS input*: scanner operation and vendor multi-setup registration are manual; everything from E57 to queryable graph—including escalation decisions and update classification—runs without human intervention, and objects the verifier cannot resolve remain explicitly **unverified** rather than being resolved by hand.

4 Methodology

This section describes the five processing stages of the framework in the order data flows through them: TLS preprocessing (4.1), geometric object decomposition with structure-remnant filtering (4.2), explicit modeling and provisional classification (4.3), multi-view VLM verification with automatic escalation (4.4), object identity matching (4.5), and confidence-gated graph upsert (4.6), followed by the structurally gated query interface (4.7) and the object-grounded place layer (4.8). Stages 4.1–4.3 form the *deterministic geometric backbone*: every random draw is seeded, so a re-run on the same host reproduces every retained cluster byte-identically (Sec. 5.2). Stages 4.4 onward are *probabilistic semantic inference*, controlled by voting, escalation, and confidence gating rather than by determinism claims. The pipeline is fully automated *after registered TLS input*: the only manual steps are operating the scanner and the vendor registration of multi-setup scans.

4.1 TLS Preprocessing

The input is a registered E57 point cloud from a survey-grade terrestrial scanner (Leica BLK360). When the export contains multiple setups, the loader applies each setup’s registered pose and merges them into a single cloud; a 70M-point six-setup scan and a pre-merged single-scan export are treated identically. The cloud is voxel-downsampled at 3 cm—dense enough to retain object detail while bounding the cost of all later stages—and normalized so that the floor lies near $z = 0$.

Structural surfaces are then removed by iterative RANSAC plane extraction: the largest plane is estimated, its inliers (4 cm threshold) are removed, and the process repeats until no plane exceeds a minimum support. Floors, walls, and ceilings—including split ceiling levels—are removed in practice within ten iterations on our scenes. RANSAC is seeded, making the extraction deterministic; the pre-removal cloud is retained as a *ceiling reference* for the remnant filter of Sec. 4.2 and for incremental re-runs.

4.2 Geometric Object Decomposition and Structure-Remnant Filtering

The structure-free cloud is partitioned into object candidates by fixed-parameter DBSCAN ($\varepsilon = 0.30$ m, 100-point minimum), followed by a *conservative giant-split*: any cluster whose horizontal footprint exceeds 2.5 m is re-clustered at a tighter $\varepsilon = 0.20$ m, and the split is accepted only if it yields at least two sub-clusters of 100+ points—separating merged neighbors without shattering genuinely large equipment. We position this stage deliberately: it is not a novel segmentation algorithm but a *reproducible decomposition designed for verification and incremental diffing*. Its value is that identical inputs yield identical clusters with stable ordering, which later stages exploit for result carry-over and change detection (Sec. 4.5). The parameter sensitivity of this choice is quantified in Sec. 5.2.

Structure-remnant filter. Real buildings violate the implicit assumption that structure is planar and objects are not. Our showroom scene has a three-level ceiling ($z = 2.48/2.18/1.95$ m): RANSAC removes each horizontal ceiling plane, but the *vertical soffit bands* at the level transitions survive removal and surface as phantom objects—thin, wall-colored slabs that a downstream verifier confidently misreads (e.g., as wall-mounted keyboards or handrails, Sec. 5.3). We therefore filter any cluster that is (i) a thin vertical planar band whose (ii) upward neighborhood in the *pre-removal* cloud terminates in a dense horizontal layer—i.e., the band hangs from a real ceiling. The topmost-layer condition prevents false positives on window sills and wall remnants, which do not have a ceiling directly above. The filter reduced the showroom scene from 45 to 38 clusters and the robot hall from 88 to 81 while leaving every retained cluster byte-identical, so previously computed verification results remain valid. Horizontal and sloped ceiling patches are out of the filter’s scope and are discussed as a residual limitation in Sec. 7.

4.3 Explicit Modeling and Provisional Classification

Each cluster is summarized by an explicit model: axis extents and centroid, yaw from the principal component of the horizontal point distribution, dominant color, and point count. These attributes are deterministic functions of the cluster; the backbone’s determinism claim covers geometry and these explicit attributes only—the provisional label below is stochastic and excluded from it. A provisional semantic label is then attached by Uni3D open-vocabulary classification over a 30-class industrial vocabulary. The provisional label is explicitly *not trusted*: our own vocabulary ablation (Sec. 5.6) shows that narrowing



Figure 2: Geometric backbone: preprocessing, plane removal, decomposition, and the structure-remnant filter (inset: ceiling-soffit phantom before/after).

the vocabulary to the deployment domain *reduces* accuracy ($66.7\% \rightarrow 33.3\%$), indicating an embedding-level domain gap that cannot be fixed by class-list curation. The provisional label serves only as VLM context and as a fallback name; every label a robot may act on must pass the verification stage below.

4.4 Multi-View VLM Verification with Automatic Escalation

Verification renders each object from four azimuths (0° , 90° , 180° , 270°) and submits each view independently to a VLM (`claude-sonnet-4-6`) under a forced tool-use schema: the model must return a type, a confidence, and a stated reason, with the object’s measured dimensions and provisional label provided as context. Free-text answers are impossible by construction, and prompts, decoding parameters, and model version are recorded with every result.

Per-view votes are combined by confidence-weighted late fusion into a fused type and a *vote share*. Three outcomes follow. If the votes are unanimous, the object is **verified**; unanimity is empirically a strong signal (94.4–100% precision on the two scenes’ confirmed ground truth, Sec. 5.3), although Sec. 5.3 also documents its failure modes. If the vote splits, the object *escalates automatically*: eight additional images—four diagonal azimuths and four zoomed close-ups—are classified, and the fusion is recomputed over all twelve votes. If the post-escalation share reaches 0.6 the object becomes **verified_escalated**; otherwise it remains **unverified**. Crucially, unresolved objects are not hidden, discarded, or resolved by a human: they stay in the graph with their failed candidate label attached, and the query interface (Sec. 4.7) is responsible for making their status impossible to ignore. Implicit robot-facing attributes (movability, openability, key-object status) are inferred once at fusion time from the winning view set.

Escalation bounds cost: only split votes pay for the wider view set (trigger rates 51–58%, Sec. 5.3), and per-call cost is $\sim \$0.009$, so a scene verifies for a few dollars (Sec. 7).

4.5 Object Identity Matching

The update problem is an *identity* problem before it is a geometry problem: a robot that plans against node IDs needs yesterday’s chair to be today’s chair. Our policy is **mint-once**: a node ID is minted when an object first enters the graph—derived from a fingerprint of its birth-time anchor (map, position, extent)—and never changes afterwards. The fingerprint is a candidate-retrieval key, not the identity itself; after minting, matching alone carries identity forward.

Matching between a new scan’s records and existing nodes runs in two passes:

1. **Spatial pass.** A record matches a node if centroids agree within 0.5 m and each extent agrees within 30% (or 0.15 m for small objects). Matches update pose and extents in place (**unchanged** / **updated**).
2. **Move pass.** Remaining records match remaining nodes of the *same type* with consistent dimensions within a 6 m search radius, producing **moved** nodes with preserved IDs. Generic types (**clutter**, **unknown**) are excluded from this pass: a generic label carries too little identity for displacement claims, and our real two-epoch data showed that admitting them fabricates 2–5 m “moves” between unrelated piles (Sec. 5.5).

Records that match nothing are minted as new nodes (**inserted**); nodes that no record claims are marked **absent** but retained with their full history, so an object that returns later resumes its original identity. Sec. 5.5 evaluates matching accuracy, identity-switch rate, and false-new/false-absent rates, including the occlusion level at which matching breaks (40% crop).

chair_040 — split base vote → automatic escalation → fused 'chair' (share 0.677, verified_escalated)

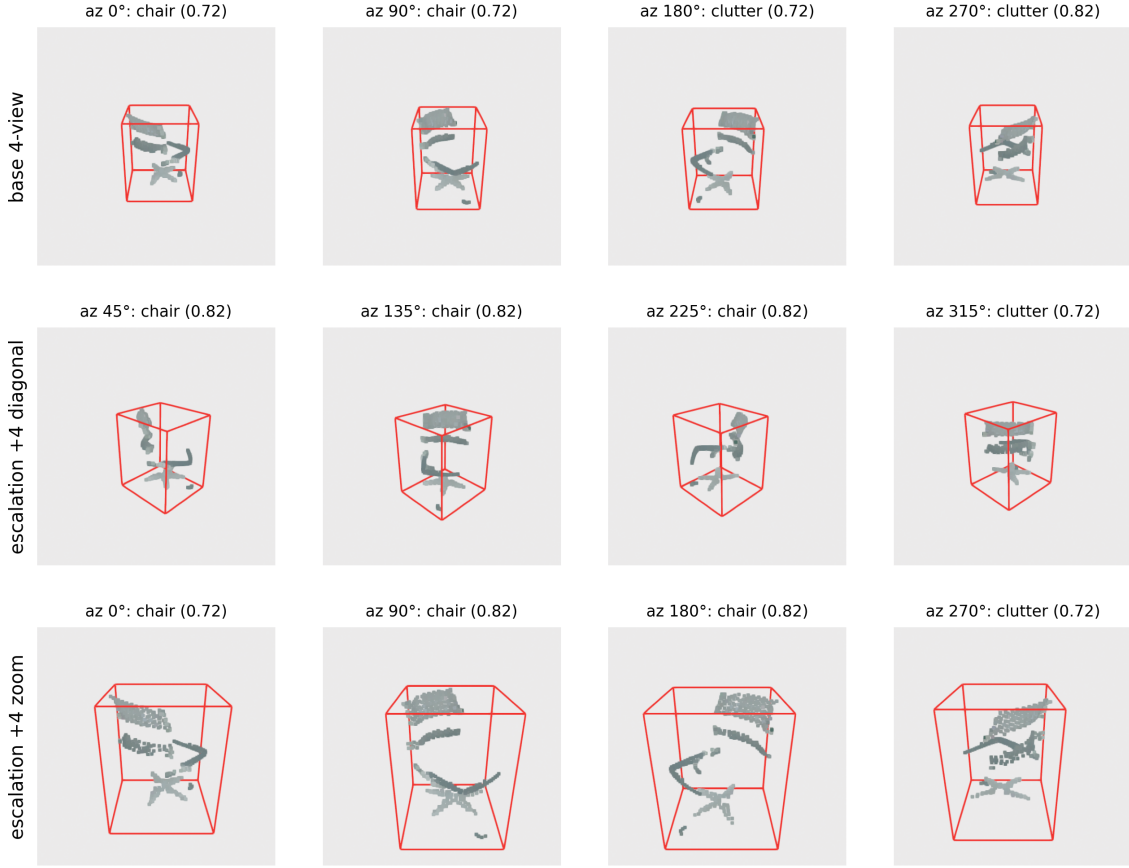


Figure 3: Multi-view verification and automatic escalation, with per-view votes, fused share, and the three terminal statuses.

4.6 Confidence-Gated Graph Upsert

Each scan epoch applies its matched record set to the knowledge graph as a node-level upsert—no rebuild. Every node carries its verification status, provenance (per-view votes, fused share, escalation flag, model version), a per-revision history of every change, and `firstSeen/lastSeen` timestamps. Edges (spatial relations recomputed from explicit models) are scoped to the map and revision. The result is exported both as a canonical JSON graph and as mapId-scoped Neo4j Cypher, so multiple sites and epochs coexist in one database. Because unchanged objects reproduce byte-identical clusters (Sec. 4.2), their verification results are carried over by content hash and a repeat scan costs *zero* VLM calls; across-epoch scans pay only for changed clusters (3.1% of full cost in our measurement, Sec. 5.5).

4.7 Query Interface with Structural Gating

The graph is queried through an LLM that receives a serialized view of the relevant nodes. The naive design—instructing the model not to assert unverified labels—fails in practice: in our benchmark the instructed model still asserted trap labels in 3 of 4 cases (Sec. 5.6). We therefore *gate by construction, not exhortation*: in the query-time view, every unverified node’s type is demoted to `unknown`, and the failed candidate label is moved to a separate `failedCandidateType` field whose semantics (“a classifier guess that did not survive verification”) are stated in the schema, not in behavioral instructions. Fig. 5 contrasts the two record states as the query interface sees them. With the gate in the data rather than the prompt, the model cannot assert an unverified label without visibly contradicting its input: the four unverified-label traps are blocked in the gating-style ablation (4/4, Sec. 5.6), while in the end-to-end benchmark one count-phrased trap still leaks through the quarantine field itself (3/4, Sec. 5.4). Traps whose label was wrongly *verified* lie beyond any status-based gate—a bound we quantify in Sec. 5.4.

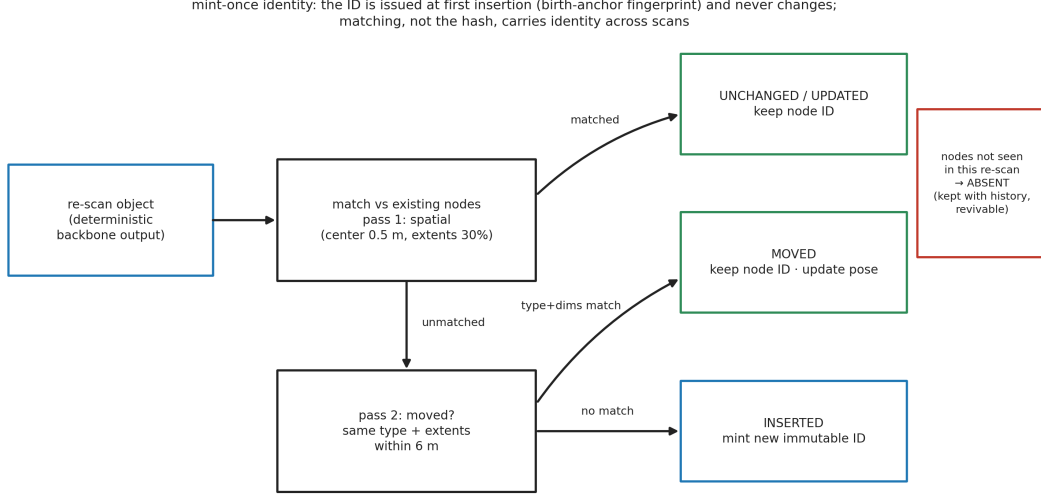


Figure 4: Identity matching and confidence-gated upsert: mint-once IDs, two-pass matching, and the five node outcomes across a re-scan.

(a) verified object record	Symbolic	
	type	control panel
	name	control_panel_007
	verified	true (VLM 4-view unanimous)
	voteShare	1.0
	Explicit	
	pose	x -2.24 y -3.54 z 1.00 m (map)
	theta	3.03 rad
	dims	1.87 x 0.43 x 2.83 m
	color	brown
	conf.	1.00
	Implicit	
	keyObject	true
	movable	false

(b) unverified record (gated)	Symbolic	
	type	unknown (gated; was 'keyboard')
	name	keyboard_046
	verified	false → demoted at query time
	voteShare	0.281
	Explicit	
	pose	x 8.59 y 4.22 z -0.58 m (map)
	theta	1.67 rad
	dims	0.58 x 0.06 x 0.51 m
	color	brown
	conf.	0.28
	Implicit	
	keyObject	false
	movable	true

Figure 5: TOSM semantic-object records (symbolic / explicit / implicit layers). (a) A verified record asserts its type. (b) An unverified record is structurally gated at query time: the type is demoted to **unknown** and the failed classifier guess is quarantined in **failedCandidateType**.

4.8 Object-Grounded Place Layer

Robot tasking refers to places (“the docking area”) as often as to objects. Following the TOSM place schema of DK-SMF [5]—boundary polygon, containment, and purpose—we add a place layer grounded in the verified object layer. Place regions are obtained automatically from the robot-facing occupancy grid map (0.05m—generated from the scan here, and equally the exploration SLAM map on deployment), following DK-SMF’s recipe: the room interior delimited by the mapped outer boundary is partitioned by SLIC superpixel segmentation [16] with the number of superpixels as the governing parameter (7 requested here), fragments below an area threshold (1.5 m²) are discarded, and each region’s centroid is the moment-based center of its cells (cf. grid-map room segmentation [17, 18]). Each object’s **isInsideOf** resolves to the region containing its anchor cell. Each place is then summarized by its *ring code*: the sequence of member objects sorted by bearing around the place centroid—restricted to verified-tier objects, so the confidence gate propagates into place semantics—and a single LLM call names the place from its ring code. Because the segmentation depends only on the map, region identity is stable across scan epochs, and the same regions re-derive names that track the room’s actual use (the machine bay: exhibition-era **machine_access_area** → storage-era **clutter_zone** → test-era **machine_operator_station**; Sec. 5.5), demonstrating that place semantics stay current without manual re-annotation. Fig. 6 shows the resulting

object-grounded place layer (T3): SLIC superpixel regions over the nav-stack grid map;
LLM-derived names from verified-object ring codes

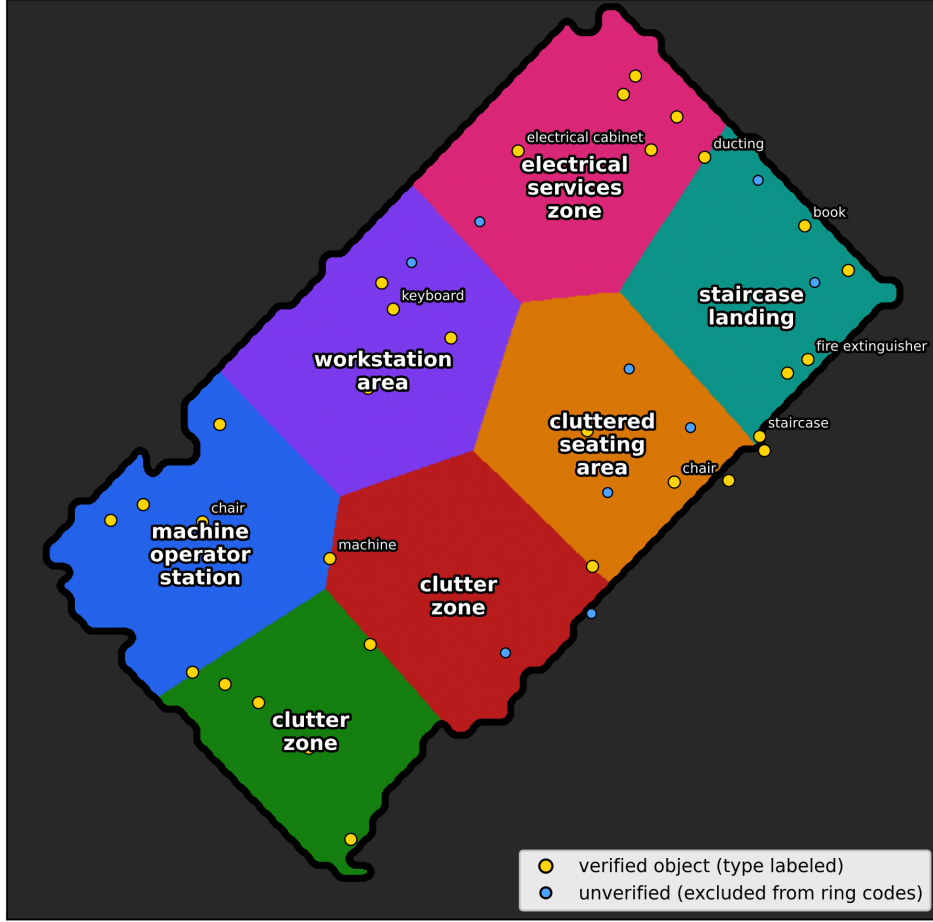


Figure 6: Object-grounded place layer at T3. SLIC superpixel regions [16] tile the room interior delimited by the nav-stack grid map’s outer boundary and carry LLM-derived names computed from verified-object ring codes; unverified objects (blue markers) do not contribute, so the confidence gate propagates into place semantics.

T3 place layer. The robot layer of TOSM is out of scope here.

5 Experiments

5.1 Setup

Scenes and epochs. All data comes from BLK360 scans of two industrial indoor environments that are, in fact, one room observed across its working life—a property we exploit deliberately. The *showroom* epoch (June 2026) captures the room in an exhibition configuration, scanned manually with the door open; its semantic target is the wall-removed export (38 objects after remnant filtering). The *robot hall* epoch (July 2026) captures the same room reorganized as a robot test hall, scanned by autonomous visibility-driven exploration (81 objects, room-scoped to 55). A third epoch (late July 2026) adds a controlled-change re-scan at 99.6% coverage (Sec. 5.5). Because the two primary epochs differ in layout, furnishing, and scan modality, they function as independent scenes for per-epoch evaluation (Secs. 5.2–5.4) while enabling a real multi-epoch update study (Sec. 5.5). Scene statistics are given in Table 3.

Ground truth. Ground truth was established in two passes: a render-based audit of every cluster, followed by confirmation from the room’s owner. Labels are graded *confirmed* (owner-verified), *plausible*, or *unknown*; accuracy is reported against confirmed GT (with lenient variants including plausible where stated), unknown objects are excluded, and structure artifacts are tracked separately. Two showroom

Table 3: Scene statistics (det sets, after structure-remnant filtering). T1 semantic processing starts from a manually wall-removed export (4.9M points) of the 46.3M-point raw scan.

	Showroom (T1)	Robot hall (T2)	Re-scan (T3)
Scan mode	manual, 1 setup	exploration	exploration (SOTA), 6 setups
Raw points	46.3M	68.6M	70.2M
Clean cloud (post-removal)	—	267K	294K
Objects (filtered)	38	81	79
Room-scoped	38	55	41
Removed planes	10	10	10

sets must be distinguished: Table 3 counts the 38-object reproducible det set, while the verification study (Sec. 5.3) uses the original 57-cluster decomposition of the same scan, of which 50 clusters are GT-scored after excluding 7 structure artifacts—yielding confirmed 36 / plausible 12 / unknown 2. For the room-scoped robot hall (55 objects), GT covers 51 clusters: confirmed 32 (lenient $n=42$) after owner corrections.

Metrics. Classification accuracy is top-1 against GT under a documented synonym-tolerant matching (e.g., *mobile manipulator* $\approx$ *robotic arm*). Verification quality is reported as per-tier precision (unanimous, escalated, unverified). For query evaluation, a *hallucination* is an unsupported atomic factual claim, and the rate is unsupported claims over all factual claims in the answer set. Update quality uses the metrics of Sec. 4.5: matching accuracy, identity-switch rate, false-new, false-absent, and unchanged stability.

Controls. All query conditions share the same LLM (claude-sonnet-4-6), system prompt, query set, decoding parameters, retry policy, and forced answer schema; conditions differ *only* in the knowledge view they receive. The LLM-only condition receives the same scene’s Stage-A object dump, so it is not information-starved—it is verification-starved.

5.2 Segmentation: Repeatability, Filtering, and Sensitivity

Repeatability. Running the full pipeline twice from the same E57—including ten-plane structure removal—reproduces every cluster byte-identically on both scenes (hash-verified). This is the property the incremental machinery builds on: identical clusters mean carryable verification results and attributable diffs.

Remnant filter. The structure-remnant filter of Sec. 4.2 removed all fourteen owner-confirmed ceiling-soffit phantoms (seven per scene: showroom 45→38, robot hall 88→81) while leaving every retained cluster unchanged. Before the filter, these phantoms did not merely inflate the object count: they produced confidently verified fictions (keyboard panels, handrails) that survived multi-view voting, since every view showed the same wall-colored slab. Detecting structure remnants *geometrically*, upstream of semantics, proved more reliable than asking the verifier to doubt itself (cf. the pose-context ablation, Sec. 5.6).

Parameter sensitivity. A 36-combination sweep (giant-split footprint $\times$ DBSCAN $\varepsilon \times$ minimum points) on both scenes shows a stable plateau at footprint 2.0–2.5 m: instance recall degrades at 3.0 m as neighbors merge (recall 0.929) and at 1.5 m as equipment shatters, while ε /min-point extremes collapse recall to 0.21. The operating point (2.5 m / 0.30 / 100) sits inside the plateau rather than on a peak. One caveat is circularity: the sweep’s reference instances are the audited operating-point decomposition, so the sweep measures stability around the operating point, not global optimality.

Conditional Point-SAM refinement. Under a pre-registered rule (adopt iff $\geq 50\%$ of merged-cluster failures separate correctly *and* label accuracy improves ≥ 5 points), Point-SAM separated 3/6 flagged merged clusters and sub-cluster labels reached 9/11 versus 5/11 for their parents (+36.4 points), so refinement is adopted—but only for *flagged merged candidates*, since automatic candidate selection also produced 2/6 false positives that fragment single objects. The T3 epoch supplied a stress case: at 99.6% coverage, inter-object gaps close and DBSCAN chains 120K points (41% of the clean cloud) into one 10.4×9.6 m mega-cluster that survives the giant-split. Refinement recovers little of the resulting diff

Table 4: Verification accuracy against confirmed GT, and per-tier precision. Accuracy rows are over confirmed GT ($n=36/32$); trigger rates and tier counts are over all GT-scored clusters (50 showroom / 51 robot hall), so, e.g., 21 unanimous + 29 split = 50.

	Showroom ($n=36$)		Robot hall ($n=32$)	
	base-4	escalated	base-4	escalated
Accuracy	75.0%	86.1%	78.1%	84.4%
Recovered / regressed	5 / 1		3 / 0	
Escalation trigger	58% (29/50)		51% (26/51)	
Unanimous precision	100% (21/21)		94.4% (of 25)	
Escalated precision	100%		88.9%	
Unverified precision	50%		40%	

inflation—Point-SAM ($K=7$ prompts) re-associates 4 of 43 absent nodes, tighter re-splitting ($\varepsilon=0.12$) 5 of 43—which is itself informative: $\sim 88\%$ of that epoch’s absences reflect genuine scene change, not segmentation artifacts (Sec. 5.5).

5.3 Verification: Fusion and Escalation

Escalation lifts confirmed-GT accuracy from 75.0% to 86.1% on the showroom and from 78.1% to 84.4% on the robot hall (lenient: 76.2% $\rightarrow$ 83.3%, $n=42$), recovering 5 and 3 split-vote objects while regressing 1 and 0 (Table 4; Fig. 7 shows the accuracy–cost trade-off at $\sim \$0.009/\text{call}$; escalation adds 232 and 208 calls per scene). The tier structure justifies gating: unanimous and escalated tiers reach 88.9–100% precision, while the unverified residue is 40–50%—exactly the content the query interface must demote. A separate check evaluates the room-scoped T1 objects in the update-study frame (Sec. 5.5), where showroom GT is transferred to the re-decomposed clusters by bidirectional point overlap; on the resulting small confirmed set ($n=8$), verified-tier precision is 100% with 7/7 unanimity.

Verification yield tracks scan quality: the manual scan leaves 47% of objects unverified versus 24% for exploration scans, and re-processing the T3 scan from a single setup (1/6 coverage) drops the verified tier from 75.6% to 56.6% on identical geometry. Coverage buys verifiability.

Error taxonomy. Four recurring error classes emerged (Fig. 8): (a) *structure-remnant phantoms*—soffit bands verified as keyboards; eliminated upstream by the remnant filter; (b) *unanimous-but-wrong*—a hanging chain unanimously verified as a door handle; an escalated keyboard the owner refuted; partition fragments as chairs; at T3, a 7.4 m wall-side structure escalation-verified as a staircase that the owner also refuted—and whose phantom label then propagated into a place name (`staircase_landing`, Sec. 4.8), showing that place naming is verifier-bounded too. Unanimity is strong evidence, not proof, and the gate inherits the verifier’s blind spots; (c) *clutter absorption*—a disassembled robot, a poster stand, and a low-reflectance floor fan (674 points) each unanimously verified as `clutter`: geometrically detected, semantically flattened; (d) *render ambiguity* $\rightarrow$ `unverified`, the intended failure mode. Classes (b) and (c) motivate reporting verification as *tiered evidence* rather than truth, and drive the structural gating design.

5.4 Knowledge-Grounded Query Evaluation

We evaluate four conditions over identically constructed per-scene query sets—48 items on the showroom, 32 on the robot hall (symbolic, explicit, scene-level, and trap questions): (1) *LLM-only* (Stage-A dump, no verification), (2) *ungated KG*, (3) *verified-only KG* (unverified nodes removed), and (4) *gated KG* (unverified retained but structurally demoted, Sec. 4.7). Each scene carries four trap questions about objects that do not exist as labeled. On the showroom all four traps target *unverified-label* phantoms (keyboard, ladder); on the robot hall two target an unverified-label phantom (refrigerator) and two an *escalated-verified* phantom (keyboard)—the latter probing the gate’s verifier-quality bound. A correct system must decline to assert the trap labels. Note that LLM-only can “block” traps trivially: with no map it tends to deny that specific objects exist at all (4/4 on the robot hall, 2/4 on the showroom), so trap blocking is diagnostic only for the grounded conditions.

On the showroom (Table 5), grounding alone (ungated) lifts accuracy from 52.1% to 81.2% but blocks no traps; removing unverified content (verified-only) blocks all four traps at a steep recall price (12

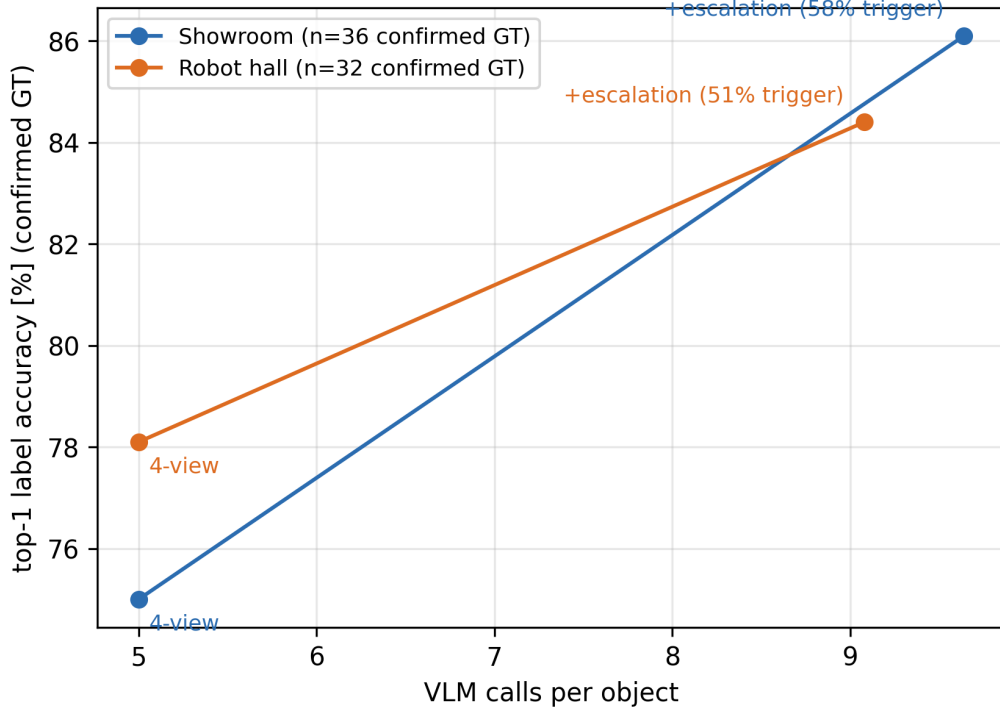


Figure 7: Escalation accuracy–cost curves for both scenes.

Table 5: Query benchmark, four conditions (accuracy / hallucination rate / traps blocked).

	LLM-only	Ungated	Verified-only	Gated
Showroom	52.1% / 20.8% / 2/4	81.2% / 18.8% / 0/4	58.3% / 16.7% / 4/4	83.3% / 16.7% / 3/4
Robot hall	50.0% / 37.5% / 4/4	56.2% / 43.8% / 0/4	46.9% / 40.6% / 2/4	59.4% / 40.6% / 2/4

abstentions, 58.3%); the gated condition is the balance point—83.3% with 3/4 traps blocked. The two leaks are instructive and mechanistically distinct. On the showroom, one *count*-phrased trap slipped through the quarantine itself: the model tallied a demoted node by reading its `failedCandidateType` field, even though it correctly denied the same object in yes/no phrasing. On the robot hall, the keyboard phantom is *escalated-verified*, so it passes any status-based gate—gating is bounded by verifier quality; we report both as properties, not anomalies. The robot hall reproduces the ordering at lower absolute levels because its end-to-end label errors (banner→refrigerator, robot→clutter) propagate into every grounded condition—an honest end-to-end number rather than a curated one.

5.5 Incremental Update Evaluation

Synthetic scenarios. Controlled edits of the room-scoped robot hall isolate matcher behavior (Fig. 10): (A) *unchanged repeat*: 55/55 nodes carried, zero VLM calls, zero identity switches; (B) *move*: 1/1 moved with ID preserved; (C) *remove+insert*: exact absent/insert classification; (D) *occlusion*: at 20% crop both affected objects still match (as updated, labels kept), at 40% matching fails into 2 false-absents + 2 false-news—the matcher’s quantified breaking point, and the motivation for overlap-based matching features in future work. On the combined-edit re-scan (move, insertion, and removal applied together), selective re-verification re-ran Stage B on only the two changed clusters: 26 of 822 VLM calls, \$0.23 versus \$7.37 for a full re-verification—3.1% of rebuild cost, and 0% for scenario A.

Real three-epoch study. The three epochs of Table 3 form a real update sequence: T1 (manual, June) → T2 (exploration, early July) → T3 (99.6%-coverage exploration with a controlled-change protocol, late July). Cross-epoch registration is fully automatic (FPFH+RANSAC then ICP; 5.1 cm RMSE for T1→T2, 4.2 cm for T3→T2). The T1→T2 diff records the room’s exhibition-to-test-hall reorganization (updated 4 / moved 3 / inserted 48 / absent 31; among the moves, a fire extinguisher displaced 3.0 m). This epoch

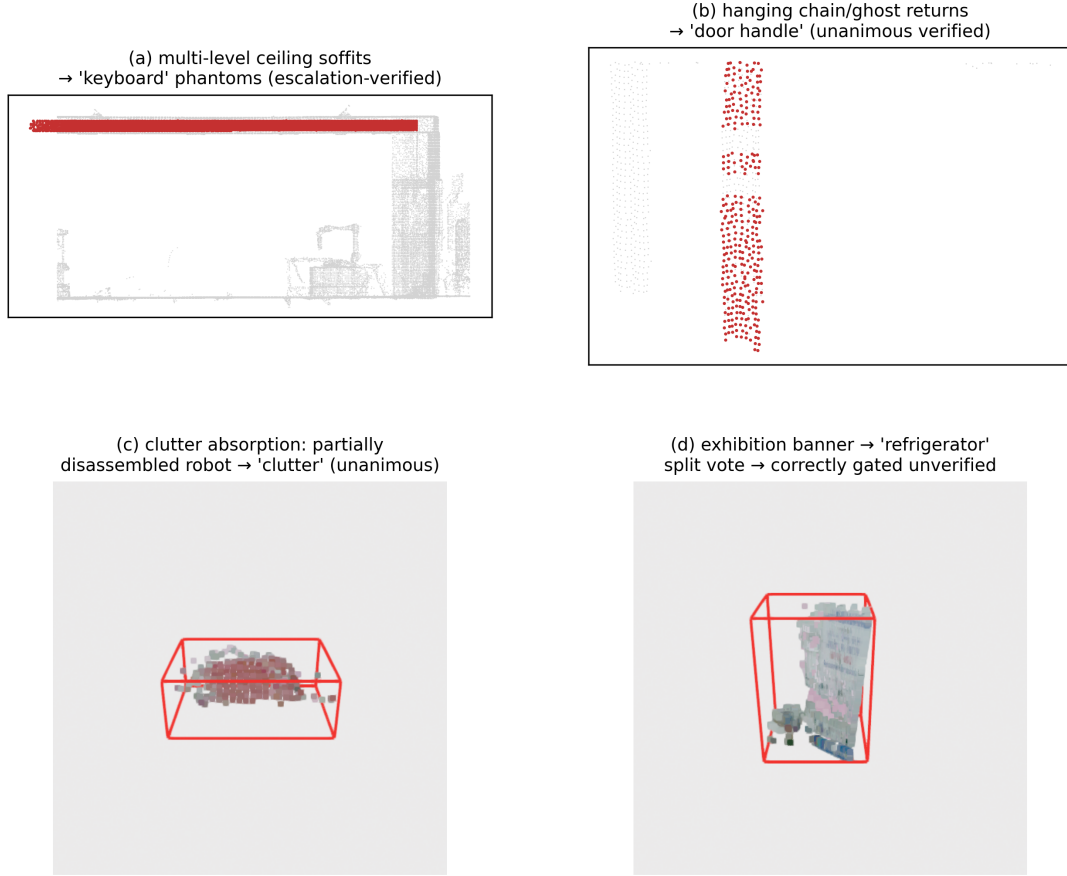


Figure 8: Verification error taxonomy: remnant phantom, unanimous-but-wrong, clutter absorption, render ambiguity.

also motivated the generic-type exclusion in the move pass: without it, clutter piles matched across 2–5 m as fictitious “moves” (9 candidates, 3 credible). The T2→T3 diff (matched 9 / moved 1 / inserted 31 / absent 45, Fig. 11) sits on top of measured natural drift: only 65% of in-room object geometry is mutually supported at 10 cm between the two scans two weeks apart—roughly a third of the room changed before any protocol was applied.

Against the controlled-change protocol, the system detected **2/2 physical changes at geometry level**: the moved chair is the epoch’s *only* moved node (0.64m, ID preserved—a node matched continuously since T1), and the introduced floor fan was correctly **inserted** but typed **clutter** (error class (c) above), giving 1/2 at type level. Identity persistence is demonstrated across all three epochs: the fire extinguisher, pump, industrial robot, and several chairs retain their minted IDs through both reorganizations. The place layer re-derives per-epoch names on the same SLIC regions (Sec. 4.8): exhibition-era `machine_access_area` and `operator_control_station` become clutter-dominated at T2 and `machine_operator_station/electrical_services_zone` at T3, at ~\$0.03 per epoch.

5.6 Ablation and Sensitivity

Escalation on/off is the base-4 column of Table 4: +11.1/+6.3 points. *Gating style*: instructing the model to respect verification status blocks 1/4 traps; structural demotion blocks 4/4 under identical prompts in this isolated comparison (in the separate end-to-end run of Table 5, one count-phrased trap leaked, 3/4)—still the single strongest argument that reliability must live in the data contract, not in instructions. *Pose context*: injecting mounting-height context as a targeted defense removed all three phantom ceiling keyboards in its pilot, but applying it to every object *reduced* escalated accuracy (86.1%→72.2%) and escalated precision (100%→72.7%)—a desk-height prior that helps ceilings hurts everything else, so it ships as a targeted defense only. *Uni3D vocabulary*: generic-30 66.7% > curated-20 48.7% > owner-18 33.3% (run-to-run variance ~5 points), confirming the embedding-level domain gap that motivates verification. *Segmentation parameters*: see Sec. 5.2.

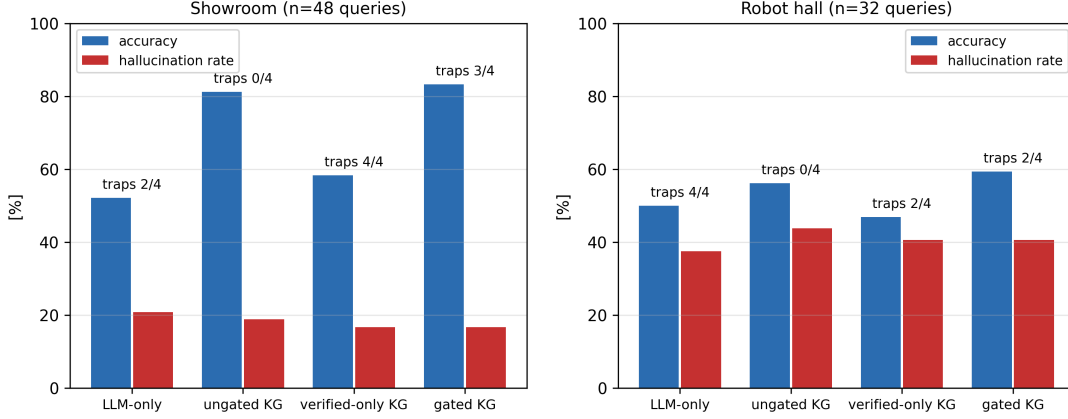


Figure 9: Four-condition query benchmark with trap-question panel.

6 Application Demonstrations

The demonstrations in this section illustrate integration surfaces; they are not claimed as contributions.

6.1 Knowledge-Grounded Dialogue

The gated query interface supports task-language interaction in which confidence is part of the answer, not metadata. Asked “is there a keyboard in the room?”, the system answers from the gated view: a verified keyboard is asserted with its location; an unverified one yields “an object at (x,y) was proposed as a keyboard by the classifier but did not pass verification”—an answer a robot can act on conservatively (route a person, lower approach speed) instead of a confident guess.

6.2 Simulation Twin Injection

Because every node carries an explicit model in the scan frame, the graph injects directly into a physics simulation. We reuse the digital-twin construction of our prior work [19] and add the semantic layer: each verified object spawns a labeled proxy (extent box or segmented cloud) in NVIDIA Isaac Sim, with verification status and provenance exposed in the scene inspector. Unverified objects spawn as neutral gray placeholders—the gate is visible in the twin.

6.3 Manipulation Readiness

The implicit layer (movability, openability, key-object status) plus verified pose and extents provide the minimum contract a manipulation planner needs for coarse feasibility screening. We state this as an interface property; manipulation experiments are out of scope.

7 Discussion

Cost and latency. VLM verification is the only paid stage. Escalation is bounded by its trigger rate (51–58% of objects), and per-call cost is $\sim \$0.009$; full verification averages $\sim \$0.09$ per object ($\$3.63$ for a 38-cluster run and $\$7.37$ for an 86-cluster run, all calls included). The incremental design changes the cost model qualitatively: a repeat scan costs $\$0$ (hash carry-over) and a changed scene costs in proportion to its change (3.1% of rebuild in our measurement). Verification is offline; no VLM call sits on a robot’s control path.

Residual unverified content. On exploration-quality scans, 24% of objects remain unverified (47% on the manual scan; 43% on the 1/6-coverage ablation)—scan quality is the dominant driver of verification yield. The framework’s position is to expose this residue, not shrink it cosmetically: unverified nodes stay queryable with demoted types, and the gate is only as good as the verifier (Sec. 5.3’s unanimous-but-wrong cases; the escalated-verified keyboard trap that survives gating on the robot hall). Verifier-bounded gating is a property users must design around, and we report it as such.

Controlled re-scan scenarios, room-scoped robot hall (55 objects; colored points = segmented clusters, thin boxes = explicit models)

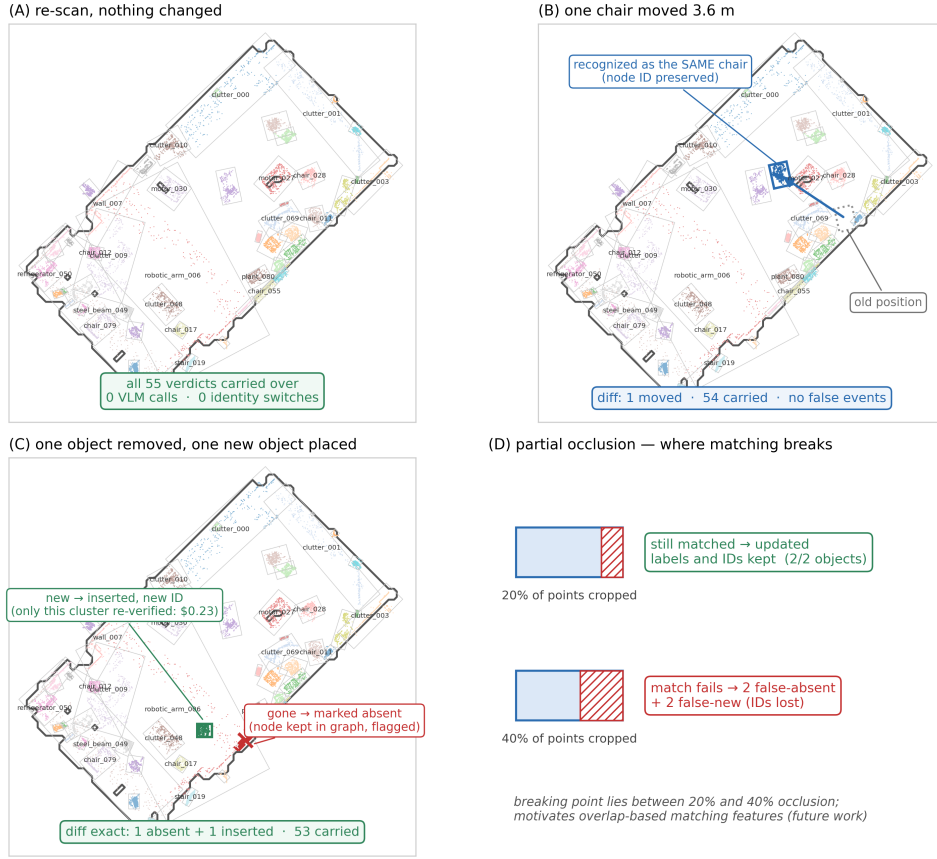


Figure 10: Controlled re-scan scenarios A–D (room-scoped robot hall, 55 objects), top-down view. Colored points are the segmented clusters, thin rotated rectangles their explicit-model footprints with node names, and the dark contour the test-room boundary; each edited object is annotated with the matcher’s decision. (A) A repeat scan with no changes carries all 55 verdicts at zero VLM cost. (B) A 3.6 m chair move is classified **moved** with its node ID preserved. (C) A removed object is marked **absent** (node kept and flagged) and a newly placed object is **inserted** under a new ID; only the insertion is re-verified (\$0.23). (D) Partial occlusion: both cropped objects still match at 20%, but at 40% matching fails into two false-absent plus two false-new pairs—the matcher’s quantified breaking point.

Small low-reflectance objects. The clutter-absorption error class—disassembled robot, poster stand, and the controlled-change floor circulator all verified as **clutter**—shows a systematic blind spot: geometrically detected, semantically flattened. Change detection still works (the circulator was correctly inserted), but type-level recall for small dark appliances needs either close-range imagery or targeted re-rendering.

Structure remnants. The soffit filter removes vertical transition bands, but horizontal and sloped ceiling patches and content seen through openings survive structure removal (six residual cases in the robot hall). These are bounded, enumerable artifacts; extending the filter to non-vertical remnants is mechanical future work.

The coverage paradox. Better scans can make density-based decomposition worse: at 99.6% coverage, gaps between adjacent objects close and DBSCAN chains 41% of the scene into one mega-cluster. Our measurements show this artifact accounts for only $\sim 12\%$ of the epoch’s absent nodes (5/43 recoverable by tighter re-splitting; 4/43 by Point-SAM refinement)—the rest is genuine change—but the effect argues for coverage-adaptive clustering or learned instance priors as the backbone’s next iteration, applied under the same flagged-candidate discipline as our Point-SAM rule.

Scope. Results come from one scanner, one building, and object-level change; structural change (walls, large fixtures) and cross-site generalization are untested. The three-epoch study mitigates but does not remove the two-scene limitation. The place layer’s regions are parameterized SLIC superpixels over the grid map: spatial cells rather than architecturally meaningful rooms, with the superpixel count fixed by hand; inferring wall- and doorway-aligned region boundaries is future work.

8 Conclusion

We presented TLS-SMF, a framework that turns registered terrestrial laser scans into a queryable, confidence-aware, and maintainable semantic knowledge graph for indoor robots. A deterministic geometric backbone makes decomposition reproducible and diffs meaningful; multi-view VLM verification with automatic escalation attaches calibrated trust to every label while retaining what it cannot resolve as explicitly unverified; and mint-once identities with two-pass matching turn re-scans into node-level upserts whose cost scales with actual change. Across two industrial scenes and a real three-epoch study of one room, the framework improved verification accuracy by up to 11 points through escalation, blocked hallucination traps by gating the data rather than instructing the model, preserved object identity across three scan epochs including a controlled physical change, and re-verified repeat scans at zero marginal VLM cost.

Future work follows the limitations surfaced by our own evaluation: extending the remnant filter to non-vertical structure artifacts, coverage-adaptive decomposition to resolve the mega-cluster paradox, close-range or re-rendered imagery for small low-reflectance objects, structural-change updating beyond the object level, automatic place segmentation to close the loop with the DK-SMF lineage, and embodied validation in which a robot plans against the graph—including its unverified residue—in the loop.

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real multi-epoch update of the same room, latest transition (rev 2 $\rightarrow$ 3)
(cross-scan registration 4.2–5.1 cm RMSE; identity carried for specific-type objects)



Figure 11: Real multi-epoch update, latest transition (T2 $\rightarrow$ T3), top-down view. Colored points are the T3 segmented clusters inside the test-room boundary; yaw-rotated node footprints are colored by diff state (matched, moved with ID preserved, inserted, absent), with names on specific-type objects. Oversized footprints of merged mega-clusters (e.g., **machine_012**, Sec. 5.6) are faded.